

Q1:

1. Draw the box plot.
2. Find a measure of central tendency. **Median = 45**
3. is the value “15” and outlier? Why? **[NO]**

- $IQR = Q3 - Q1 = 60 - 40 = 20$
- Lower Bound = $Q1 - 1.5 \times IQR = 40 - 30 = 10$
- Upper Bound = $Q3 + 1.5 \times IQR = 60 + 30 = 90$

4. Direction Skewness: **right / positive skewed**

5. Calculate the coefficient of skewness

بالمعادلة هتبقى قيم تقريبية ف استخدمت بايثون

```
mean_value = np.mean(data)
median_value = np.median(data)
std_value = np.std(data, ddof=0) |
skewness = (3 * (mean_value - median_value)) / std_value
```

- **Mean: 55**
- **Median: 45**
- **Standard Deviation: 22.47**
- **Skewness Coefficient: 1.33**

Comment: **right-skewness is confirmed due to the positive value.**

6. Is it a difficult exam? Why? **YES**, because:

- The median is 45 (**below the midpoint**).

